

Lab03_Pan

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Criação das variáveis “flights”, “airlines” e “airports”

```
library(readr)
library(dplyr)
```

Warning: pacote 'dplyr' foi compilado no R versão 4.5.3

Anexando pacote: 'dplyr'

Os seguintes objetos são mascarados por 'package:stats':

filter, lag

Os seguintes objetos são mascarados por 'package:base':

intersect, setdiff, setequal, union

```
library(stringr)

airlines <- read_csv("airlines.csv")
airports <- read_csv(
  "airports.csv",
  col_types = cols_only(
    IATA_CODE = col_character(),
    CITY = col_character(),
    STATE = col_character(),
    LATITUDE = col_double(),
    LONGITUDE = col_double()
  )
)
```

```

)

flight_processing <- function(chunk, pos) {
  chunk %>%
    na.omit() %>%
    filter(!str_starts(as.character(DESTINATION_AIRPORT), "1")) %>%
    group_by(DESTINATION_AIRPORT) %>%
    summarise(
      soma_atraso = sum(ARRIVAL_DELAY),
      total_voos = n(),
      .groups = "drop"
    )
}

flights <- read_csv_chunked(file = "flights.csv.zip",
                           callback = DataFrameCallback$new(flight_processing),
                           chunk_size = 1000000,
                           col_types = cols_only(
                             DESTINATION_AIRPORT = col_character(),
                             ARRIVAL_DELAY = col_double()
                           ))

```

Contando aeroportos por Estado

```

aeroportos_ausentes <- flights %>%
  anti_join(airports, by = c("DESTINATION_AIRPORT" = "IATA_CODE")) %>%
  pull(DESTINATION_AIRPORT)

flights <- flights %>%
  left_join(airports %>% select(IATA_CODE, CITY, STATE),
            by = c("DESTINATION_AIRPORT" = "IATA_CODE"))

airports_by_state <- airports %>%
  filter(!is.na(STATE)) %>%
  group_by(STATE) %>%
  summarise(
    total_aeroportos = n(),
    total_cidades = n_distinct(CITY)
  ) %>%
  arrange(desc(total_aeroportos))

```

```
airports_by_state
```

```
# A tibble: 54 x 3
  STATE total_aeroportos total_cidades
  <chr>          <int>          <int>
1 TX             24             23
2 CA             22             21
3 AK             19             19
4 FL             17             17
5 MI             15             15
6 NY             14             13
7 CO             10             10
8 MN              8              8
9 MT              8              8
10 NC             8              8
# i 44 more rows
```

```
library(webshot2)
```

Warning: pacote 'webshot2' foi compilado no R versão 4.5.3

```
library(tinytex)
library(leaflet)
```

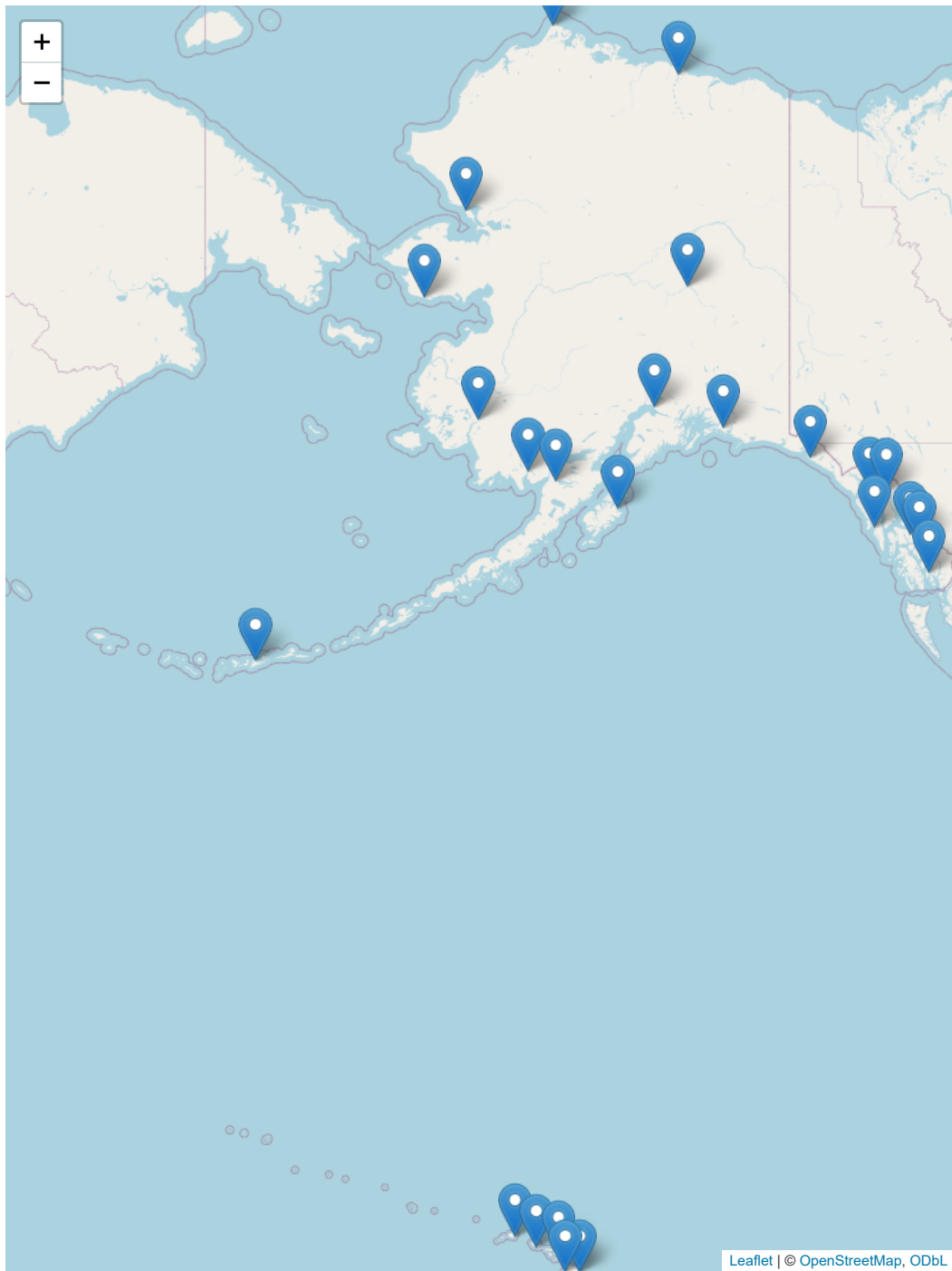
Warning: pacote 'leaflet' foi compilado no R versão 4.5.3

Criação de um mapa com o local de cada aeroporto

```
leaflet(data = airports) %>%
  addTiles() %>%
  addMarkers(
    lng = ~LONGITUDE,
    lat = ~LATITUDE,
    popup = ~IATA_CODE
  )
```

Warning in validateCoords(lng, lat, funcName): Data contains 3 rows with either missing or invalid lat/lon values and will be ignored

file:///C:/Users/artur/AppData/Local/Temp/RtmpMHQoN0/file55cc71a51dc/widget55ccc015846.html ;



```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-09-01 02:52:03 -03"
```